

F M S 11 - 4**SUPPLEMENT FOR
SPECIAL TAKEOFF AND
LANDING OPERATIONS**

NOTE Due to the nature of its content, this supplement is divided into separate, related subsections that first present General Data applicable to all Special Takeoff and Landing Operations and then specific data applicable to approved types of operation.

A. GENERAL DATA**1. GENERAL**

This supplement shall be attached to the BO 105 Flight Manual (Section 11) when the VTOL Retrofit Kit has been installed.

System/Equipment Designation	Part No.	Effectivity
VTOL Retrofit Kit	105-80037	All
NOTE Helicopters S/N 0001-0450 must have SB 60-37 (ram air ventilation) installed.		

The information contained herein supplements the information of the basic Flight Manual; for limitations, procedures, and performance data not contained in this supplement, refer to the basic Flight Manual.

NOTE For approving authorities and original issue date of this supplement, refer to the Log of Supplements.

1.1 INTRODUCTION

Although the BO 105 helicopter basically is certified for operation according to FAR 27, this supplement provides standardized takeoff and landing procedures and performance data which are certified according to FAR 29, Category A operations.

This supplement only applies to helicopters having main transmission ZF FS72E (Retrofit kit, P/N 105-80037) installed.

The information presented in this supplement is required for the following approved types of operations:

- Elevated or restricted helipad (VTOL)

Presented performance data apply to the minimum requirements of FAR 29, subpart B.

1.2 ABBREVIATIONS

AHE	- Above Helipad Elevation
LDP	- Landing Decision Point
MAT	- Mass/Altitude/Temperature
TDP	- Takeoff Decision Point
TOGM	- Takeoff Gross Mass
V _{TOSS}	- Takeoff Safety Speed
VTOL	- Vertical Takeoff and Landing

1.3 DEFINITIONS

Takeoff Decision Point (TDP) is the last point in the takeoff path at which, in the event of failure of one engine, a rejected takeoff is assured within the distances determined according to the procedure used, and is the first point at which a continued takeoff is assured.

If one engine fails prior to TDP the takeoff must be rejected.

If one engine fails after TDP the takeoff may be continued.

Landing Decision Point (LDP) is the only point on the landing profile at which, in the event of failure of one engine, the combination of height and airspeed permits the decision to proceed with the initiated landing or to accelerate to takeoff safety speed (V_{TOSS}) and to accomplish a safe climb.

If one engine fails prior to LDP the pilot may elect to go-around or continue the landing approach.

If one engine fails after LDP the landing must be continued.

Takeoff distance required is the horizontal distance along the takeoff path required to attain and remain at least 35 feet above the takeoff surface, to attain and maintain a speed of at least V_{TOSS}, and to establish a positive rate of climb, assuming critical engine failure recognition occurs at the takeoff decision point.

Takeoff Segment I Distance is the horizontal distance necessary to climb with V_{TOSS} from the end of the takeoff distance to 200 ft AGL.

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SUPPLEMENT FOR
EQUIVALENT CATEGORY A OPERATIONS
(VTOL ONLY)

NOTE Due to the nature of its content, this supplement is divided into separate, related subsections that first present General Data applicable to all Equivalent Category A Operations and then specific data applicable to approved types of operation.

A. GENERAL DATA

1. GENERAL

This supplement shall be attached to the BO 105 Flight Manual (Section 11) when the VTOL Retrofit Kit has been installed.

System/Equipment Designation	Part No.	Effectivity
VTOL Retrofit Kit	105-80037	All
NOTE • Helicopters S/N 0001-0450 must have SB 60-37 (ram air ventilation) installed. • The Engine Fire Extinguishing System (FMS 10-25) and Retrofit Low Fuel Warning System (P/N 105-61020) must be installed and operational.5		

The information contained herein supplements the information of the basic Flight Manual; for limitations, procedures, and performance data not contained in this supplement, refer to the basic Flight Manual.

NOTE For approving authorities and original issue date of this supplement, refer to the Log of Supplements.

1.1 INTRODUCTION

Although the BO 105 helicopter basically is certified for operation according to FAR 27, this supplement provides standardized takeoff and landing procedures and performance data which are certified according to FAR 29, Category A operations.